

midWife: Enabling Open-Ended Planning via Socratic Dialogue with Large Language Models

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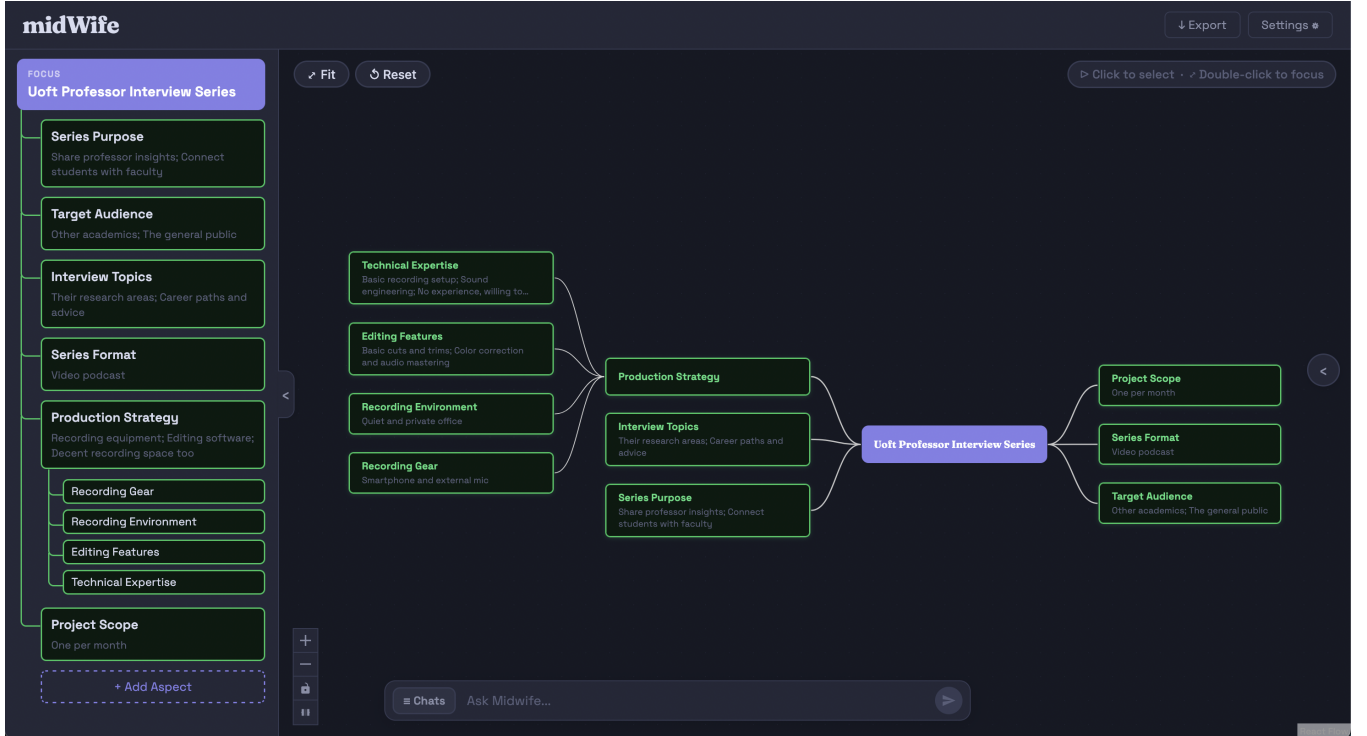


Figure 1: The MidWife discourse canvas. A user’s planning objective is decomposed into AI-generated Aspects — focused Socratic questions across key dimensions. Users answer via suggestion chips or free text; the tree expands recursively on elaboration. A per-node chat assistant (centre) guides the user to a concrete answer. The synthesis panel (right) distils the full discourse into actionable summaries adapted to the selected planning mode.

Abstract

Large Language Models (LLMs) have enabled users to engage in planning, reasoning and sensemaking with a “thought partner” of sorts, allowing open-ended discussions on nuanced subjects. This mirrors the idea of a *Socratic interlocutor* asking probing questions — serving to help users formalize their thinking, flesh out new ideas, and even find holes in their planning. However, most LLM platforms are limited to one-dimensional chat interfaces that cannot adjust to the scope of a dynamic, ever-growing discourse, thereby limiting the ability of a user to build a map of their thinking. This reflects an inherent limitation in the affordances offered by a linear

conversational interface. We present *midWife*¹: an interactive system that encourages thinking by iteratively soliciting the user’s intent, objective and/or nascent thoughts, probing with revelatory questions to concretize user-indicated notions, and visualizing this discourse as a map of thinking in the form of a node-link diagram connecting a central Thesis with its many Aspects. At each iteration, the user may (i) solicit more probing questions conditioned on a specific Aspect (if they are unsure of what to explore next); or (ii) manually add one or more Aspects at any point of the map (if they already know what to explore). Parallely, a Plan is generated by midWife in rich Markdown, updated incrementally to reflect the growing map of thinking. A persistent chat interface allows users

¹The name is a direct tribute to the Socratic metaphor of *maieutics* [12]: just as midwives assist in giving birth to children, he described his Socratic method as helping others give birth to wisdom and ideas. midWife plays the same role.

to ask questions, request changes, and steer the Plan toward their focus areas.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Natural language interfaces*; • **Computing methodologies** → Natural language processing.

Keywords

Socratic dialogue, discourse trees, structured thinking, LLM-assisted planning, mixed-initiative interaction, sensemaking

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1 Introduction

Planning complex, multi-dimensional tasks is an increasingly common yet poorly supported use case for large language models. Whether deciding how to change careers, organizing a major event, or working through a research hypothesis, users routinely turn to LLM-powered chat assistants as thinking partners [3]. Yet every major LLM interface organizes interaction as a flat, scrollable chat thread — a one-dimensional medium that cannot represent the branching, recursive structure of real deliberation [7]. A user who spends hours in a chat session planning a complex project has no spatial overview of what has been considered, no ability to navigate across parallel concerns, and no persistent artifact to return to and refine. Doing so with text-heavy media quickly becomes clunky; further, a user who wishes to explore a tangent cannot do so without polluting the shared context and potentially derailing the main thread of discussion. The *scope* of the discourse outgrows what the *interface* can hold.

The Socratic Method has long provided an answer to this kind of open-ended inquiry. Rather than providing answers, a skilled interlocutor asks focused questions that surface hidden assumptions, expose logical gaps, and map the conceptual space of a problem before any commitment is made [14, 18]. Socrates himself described this role as that of an “intellectual midwife”: whereas midwives help give birth to children, his questioning helped give birth to wisdom and ideas in others [12]. Applied to interactive systems, this lineage produced argument mapping tools rooted in Rittel and Kunz’s IBIS [6, 13], and more recently AI-powered dialectical systems [2, 17]. What remains missing is a single system that combines *spatial discourse structure* with *AI-as-Socratic-interlocutor* — one where questioning drives the growth of a persistent, navigable tree of thinking.

We present **midWife**: an interactive web application for open-ended planning via Socratic dialogue. Users state a planning objective and optionally select a *mode* (logistics, brainstorming, creative, decision, research, and more). midWife responds with a set of *Aspects* — targeted Socratic questions spanning key dimensions of the objective — rendered as nodes in a branching discourse tree. Answering an Aspect (via suggestion chips or free text) unlocks

Elaboration, which generates child Aspects conditioned on the parent’s answer and full ancestry context. A per-node chat assistant guides users through difficult questions, surfacing new Aspects or reframing questions as needed. As the tree grows, a *Synthesis Panel* distills the full discourse into structured, actionable output — timelines, task lists, open questions — adapted to the selected planning mode. The entire tree is a persistent, resumable artifact.

This paper makes the following contributions:

- **midWife**: a working prototype of LLM-powered Socratic planning via interactive discourse trees, built with a React + ReactFlow frontend and a FastAPI backend.
- A **mode-aware questioning engine** supporting nine distinct planning contexts, each modulating the AI’s Socratic questioning strategy.
- A **mixed-initiative interaction model** in which user and AI jointly construct, navigate, and refine the discourse tree [4].
- **Design insights** from iterative prototype development, including a proposed evaluation plan for future work.

2 Related Work

2.1 AI-Augmented Sensemaking and Planning Tools

A wave of systems published at UIST and CHI since 2022 have directly addressed the limitation of linear chatbot interfaces for complex information work. Sensecape [16] provides an infinite canvas where users interact with an LLM through multilevel abstraction, moving fluidly between information foraging and sensemaking loops. Selenite [9] solves the “cold-start” problem in unfamiliar decision domains by generating structured overviews of options and criteria upfront. Luminate [15] prevents premature convergence by generating responses that span a prompt’s full design space. Graphologue [5] converts LLM responses into interactive node-link diagrams in real time. Most closely aligned is Human-AI Deliberation [10], which engages users in dimension-level opinion exchange and iterative discussion rather than static recommendations. Commercially, tools such as Whimsical AI and Miro AI take a simpler approach: a single prompt generates a complete mind-map, leaving the user as a passive recipient rather than a co-constructor.

A common thread across these systems is that AI produces *content*: information, structure, synthesis. It organises and generates; it does not interrogate. midWife’s core departure is that its primary AI role is to *ask*, not to tell.

2.2 Socratic and Dialectical AI Interaction

Tanprasert et al. [17] demonstrated that LLM-powered debate chatbots most effectively induced critical thinking in users engaging with online video content. Chiang et al. [2] showed that interactive devil’s advocate agents improved appropriate reliance and decision accuracy by catalyzing extended discussion. At the model level, work on Socratic prompt engineering [1] and fine-tuned Socratic models [8] validates that LLMs can reliably ask guiding questions rather than give answers. Lee et al. [7] provide an empirical motivation for midWife’s design: higher AI confidence correlates with *less* critical engagement — underscoring the need for AI systems that provoke thought rather than satisfy it. Collins et al. [3] frame

the aspirational target as AI “thought partners” that think *with* humans.

A shared limitation across this stream is structural: these systems operate within flat, linear chat interfaces. The dialogue may be dialectical, but the representation is one-dimensional. midWife embeds Socratic questioning inside a persistent, spatial discourse tree — giving the dialogue a structure that users can navigate, revisit, and build upon.

2.3 Non-Linear Conversation and Argumentation

The intellectual lineage of structured discourse tools begins with Kunz and Rittel’s IBIS [6], which defined a grammar for structuring deliberation around Issues, Positions, and Arguments, motivated by the observation that ill-structured “wicked problems” [13] demand argumentative processes rather than algorithmic resolution. Recent work has begun augmenting this lineage with LLMs. VISAR [20] supports argumentative writing through hierarchical goal brainstorming and LLM-powered argumentation sparks within a tree-structured outline. AMQuestioner [11] presents Socratic questions over automatically extracted argument maps to guide critical exploration of online claims. Wang et al. [19] provide direct empirical support for midWife’s design choice: node-tree interfaces outperformed chatbots on decision-making and brainstorming tasks. Horvitz’s mixed-initiative principles [4] supply the interaction design framework for managing when the AI should take initiative versus deferring to the user.

Despite this progress, these systems use AI primarily for *content generation*: surfacing counterarguments, labelling argument types, retrieving evidence. None positions the AI as a *Socratic facilitator* whose primary goal is to help the user think more deeply through structured questioning.

The three threads described above are active and productive, yet no existing system integrates all three. Table 1 characterizes the landscape. Sensemaking tools provide spatial or hierarchical structure but the AI generates *content*, not questions. Dialectical AI systems challenge and interrogate but do so within flat, linear interfaces. Argumentation tools provide structured discourse representations but use AI only for content generation, not Socratic facilitation. midWife’s contribution is the synthesis: a tree-structured dialogue interface where the AI acts as a Socratic interlocutor, combining the spatial structure of Sensecape and Graphologue, the dialectical AI role of Human-AI Deliberation and the debate chatbot literature, and the argument grammar of IBIS and Toulmin — extended with mode-aware questioning across nine distinct planning contexts.

3 MIDWIFE

3.1 Motivating Scenario

Consider Priya, a software engineer contemplating a move abroad for a new career opportunity. She opens a traditional chat assistant and types her situation. The assistant replies with a few high-level considerations — finances, visa, lifestyle — but the exchange is flat: each message buries the last, tangents derail the main thread, and after twenty exchanges Priya has no spatial sense of what she has covered or what remains open.

Table 1: Positioning of midWife relative to representative prior systems. ✓ = fully present, ~ = partial, × = absent.

System	Spatial tree	Socratic AI	Mode-aware
Sensecape [16]	✓	×	×
Human-AI Deliberation [10]	×	~	×
Debate Chatbots [17]	×	✓	×
VISAR [20]	✓	~	×
AMQuestioner [11]	✓	~	×
midWife (ours)	✓	✓	✓

With midWife, Priya types: “I’m thinking of moving to Berlin for a job opportunity” and selects the *Decision* mode. midWife generates five Aspects: *Finances*, *Visa & Logistics*, *Career Fit*, *Lifestyle*, and *Family & Social*. Each appears as a node on a radial discourse canvas. She clicks *Finances*: midWife asks, “How long could you sustain yourself financially if you spent a few months settling in before starting?” She types her answer. She clicks *Elaborate* — midWife generates three child Aspects: *Cost of Living*, *Savings Buffer*, and *Currency Risk*. She navigates back up and clicks *Career Fit*, opening the per-node chat assistant. She is unsure how to think about this angle; the assistant asks, “What would you want to have achieved professionally in two years that your current role does not allow?” — guiding her to a concrete answer without providing one. After exploring four of the five top-level Aspects, she clicks *Synthesize*. midWife produces a structured Decision Panel: an overview, a list of open questions, a recommendation summary, and actionable next steps. Priya did not *receive* a plan — she *built* one.

3.2 Design Goals

D1. Visualize Discourse as a Dynamic Map of Thinking. A linear chat transcript cannot externalise the structure of multi-dimensional thinking. The interface should render the growing discourse as a spatial, navigable artefact — one that accumulates a visible record of what has been considered, decided, and left open, and that remains editable by the user throughout.

D2. Probe, Rather Than Answer, on the User’s Behalf. Many contemporary mind-mapping and planning tools offload the thinking away from the user entirely: the user provides an initial intent and is presented with a complete plan, map, or document generated by the AI. While efficient, this pattern significantly diminishes the user’s *sense of ownership* over the produced artefact — the final outcome does not carry their fingerprint. We take a fundamentally different approach. midWife uses the LLM exclusively to: (i) ask probing questions; (ii) uncover hidden assumptions; (iii) help the user verbalize inchoate ideas through focused follow-ups; and (iv) offer concrete suggestions only when the user is navigating an unfamiliar domain. The *onus* of deciding, committing, and directing always rests with the user.

D3. Model Thinking as a Mixed-Initiative Dialogue. Effective deliberation is neither fully user-driven nor fully AI-driven; it is a back-and-forth in which each party contributes what the other cannot easily do alone. We therefore designed midWife so that *every significant action* can be initiated by either the user or the AI. New Aspects can be AI-generated (via elaboration) or user-created

(via the Navigator Pane or manual node creation). Questions can be AI-authored or user-edited. The Synthesis Panel is initially generated by the AI based on the completed discourse, but the user retains full executive control: they can edit, delete, or redirect any section of the plan via natural language chat. This mirrors the core principles articulated by Horvitz [4]: the system should scope its initiative to the user's current context, offer safety nets (e.g. editable suggestions), and time AI interventions to moments when the user is most likely to benefit from them.

D4. Treat the Discourse as a Persistent Artefact. A planning session should not vanish when the browser closes. The Discourse tree and its synthesised Plan should be resumable, allowing users to return to, continue, and revise their thinking over time rather than starting from scratch each session.

3.3 The Discourse Tree Model

Each session in midWife is called a **Discourse** — a structured planning session centred around a **Thesis**: the user's objective, intent, topic of interest, or open question. Before the Discourse begins, the user specifies a **Mode of Help**: one or more of Logistics, Brainstorm, Creative, Problem-Solving, Decision, Research, Reflect, Set Goals, or Learn, which shapes the AI's questioning strategy throughout.

The fundamental unit of the Discourse is an **Aspect**: a dimension of the Thesis that warrants exploration. Each Aspect is a node in the tree carrying four fields: (i) a short *label* (at most three words, e.g. *Budget*, *Career Fit*); (ii) a *Socratic question* — a focused, beginner-friendly prompt to help the user think through that dimension; (iii) a set of three to five pre-generated *suggestion chips*; and (iv) the user's *answer*, once provided. Answering an Aspect unlocks *Elaboration*: midWife generates child Aspects whose questions are conditioned on the full *ancestry path* — every aspect/question/answer triple from the root down to the current node. This ensures deeper questions are meaningfully scoped to what the user has already decided.

Recontextualization addresses a subtle issue that emerges as the discourse deepens: an ancestor's short label may no longer accurately reflect the scope of its subtree. The system periodically re-evaluates ancestor labels against their children's content and proposes updated labels, keeping the map legible as the discourse grows.

3.4 User Interface & Features

After the user submits their Thesis and Mode of Help, the interface transitions to the main Discourse page, consisting of three panels: (i) the **Navigator Pane** on the left, a structured list of all Aspects and the primary surface for manual tree editing; (ii) the **Discourse Canvas** at the centre, a node-link diagram updated in real time as the Discourse grows; and (iii) the **Plan Pane** on the right, a synthesised Markdown document derived from the current Discourse state.

3.4.1 Interview & Answering. For each *Aspect* in the Discourse, midWife engages the user in an **Interview**: a focused, one-at-a-time question session. The user is presented with the Socratic question for the selected Aspect alongside up to five pre-filled suggestion chips. They may select one or more chips, add detail in a free-text box, or type a fully custom answer. If they are uncertain how to approach the question, they may choose *Chat about this*, which

opens the **Chat Interface** anchored to that specific node. Here, the LLM helps the user articulate their requirements, resolve ambiguity, and arrive at either a pre-specified option or a custom answer of their own. Critically, the chat assistant does not answer the question *for* the user: it asks, clarifies, and guides until the user reaches their own conclusion.

3.4.2 Mode-Aware Questioning Engine. midWife supports nine planning modes — Logistics, Brainstorm, Creative, Problem-Solving, Decision, Research, Reflect, Set Goals, and Learn — each of which injects a distinct instruction paragraph into the LLM's system prompt, redefining what constitutes a relevant Socratic question for that context. Users may select a single mode or combine multiple for hybrid sessions.

3.4.3 Navigator Pane. The Navigator Pane provides a structured list view of the Discourse, organised around the concept of a **Focus Node**. By default the root Thesis is the Focus Node, and the Pane displays it alongside up to its grandchildren — giving users a bounded, readable slice of the tree at any one time. Users may double-click any Aspect to re-focus: the Canvas recentres on that node and the Pane narrows to its local subtree, allowing deep branches to be explored without the surrounding tree becoming a distraction.

Within the Navigator Pane, users can add new Aspect nodes to any existing node or to the Thesis itself, edit labels, delete nodes, and move a node to become the child of a different parent if they feel the structure better reflects their thinking. Nodes can also be expanded by the AI: each Aspect card carries an **Explore** button that instructs midWife to generate additional directions to investigate under that Aspect, surfacing branches the user may not have considered.

3.4.4 Plan Pane. While the Discourse Canvas captures the *structure* of a user's thinking, the Plan Pane synthesises its *implications*. It surfaces information that the canvas does not directly represent: concrete action items and todos derived from answered Aspects, a timeline if the user's objective has a temporal dimension, open questions that remain unresolved in the tree, and risks or considerations that are still unaddressed. All of this is rendered as a rich Markdown document, updated on demand as the Discourse grows.

Users steer the Plan through the chat interface at the bottom of the Discourse Canvas. Each requested edit — rephrasing a section, adding a new consideration, restructuring the timeline — is first shown as a **diff preview**; the user can accept the proposed changes or dismiss them and keep the current Plan. This accept/reject flow ensures the Plan remains under the user's control rather than being silently overwritten by the AI.

3.5 Mixed-Initiative Interaction

midWife distributes initiative between user and AI across every layer of the interaction [4]. The AI generates Aspects, pre-fills suggestion chips, proposes recontextualization labels, and synthesises the Plan; the user answers, elaborates, manually edits the tree, and steers the Plan via chat.

A key design decision is scoping the chat assistant to individual nodes rather than providing a single global chatbot. Each chat session is initialised with the full context of the selected node — its label, Socratic question, and the complete ancestry path of prior answers — and is given one directive: help the user reach a concrete

answer to *this* question. The assistant will reframe the question if the user finds it irrelevant, surface a new Aspect if a genuinely new concern emerges, and extract a suggested answer as soon as the conversation converges. Each node’s chat history is preserved independently, allowing users to track the reasoning behind any answer and return to prior conversations without losing context. This scoping prevents context bleed across unrelated branches and ensures the chat always serves the discourse tree rather than replacing it.

4 Implementation

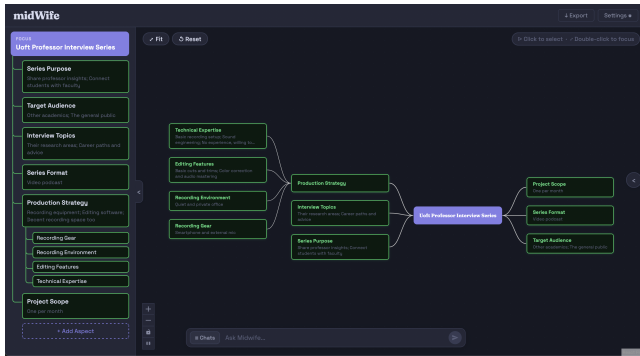


Figure 2: MidWife system architecture. The React frontend communicates with a FastAPI backend over a REST API. Claude Haiku serves as the primary LLM; Gemini models provide a cascading fallback. Session state is maintained server-side and synced to browser localStorage, enabling resumable sessions without a database.

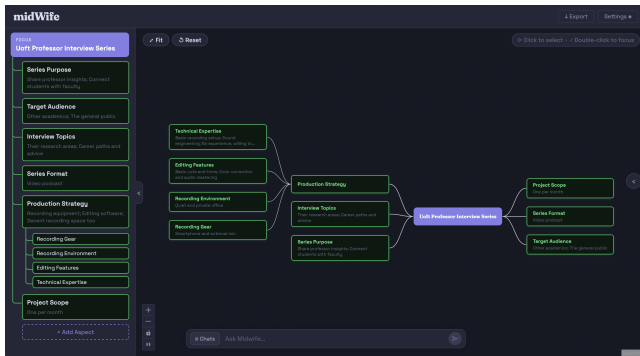


Figure 3: MidWife in use. (a) A planning session in progress: answered Aspect nodes (dark fill) branch into revealed children; the selected node’s chat assistant is open. (b) The synthesis panel generated from the completed discourse, showing mode-specific tabs (Overview, Timeline, Tasks & Resources, Open Questions).

midWife is built on a React + Vite frontend and a FastAPI backend (Figure 2). The discourse canvas is implemented with ReactFlow, which provides native pan, zoom, and node-selection at no interaction cost to the user. Tree layout is handled by Dagre, which

positions nodes automatically as the tree grows — left and right children are assigned based on edge source handle, ensuring the mindmap-style branching remains spatially consistent regardless of tree depth or width. Layout constants (rank separation, node separation) were tuned iteratively so that a fully elaborated tree with 20–30 nodes remains readable without manual rearrangement.

The AI pipeline uses Claude Haiku 4.5 as the primary model, chosen for its low latency and reliable structured-JSON output — all Aspect labels, Socratic questions, and suggestion chips arrive in a single round-trip, keeping the interview feel responsive. A cascading fallback to Gemini models provides graceful degradation under rate limits, transparent to the user. Three prompt engineering decisions directly shape interaction quality: existing node labels are injected as a “do not repeat” constraint to prevent duplicate Aspects; the full ancestry is included in every elaboration call, causing questions to deepen with the tree; and each planning mode installs a distinct questioning persona, making the same interaction model feel meaningfully different across contexts. When a user manually adds a node by label only, the system synthesises a matching Socratic question from its ancestry, keeping user-authored nodes consistent with AI-generated ones.

5 Discussion

5.1 Design Insights

Building midWife surfaced several non-obvious design lessons. **Mode-awareness emerged from necessity:** early prototyping revealed that a single prompt style produced planning-flavoured questions even for reflection and goal-setting tasks. Users asking “what do I want from my career?” were met with budget and timeline questions. The nine-mode system was a direct response, and the difference in question quality was immediately perceptible.

Per-node chat scoping proved to be the most consequential single design decision. Early iterations offered a global chat sidebar, but users may quickly drift away from specific Aspects into open-ended conversation, causing the discourse tree to stall. Anchoring each chat session to one node — and giving it one directive — kept the conversation in service of the tree rather than replacing it.

The Plan Pane functions as a goal state. In open-ended planning sessions, users may not know what “finished” looks like. Observing the Plan populate incrementally may give users a concrete sense of progress and a natural stopping condition, in a way that a growing tree alone does not. The panel effectively reframes the tree-building activity as working toward a deliverable. However, auto-refreshing the Plan on every answered question introduces a potential tension: users who have grown attached to a particular phrasing may find subsequent refreshes disorienting. Keeping the plan current and keeping it stable pull in opposite directions — a trade-off without a clean resolution.

Recontextualization is difficult to surface. The system re-evaluates ancestor labels as branches deepen, but users may not notice proposed relabellings without a stronger affordance drawing attention to them. This feature has real value — a node labelled “Budget” at depth one may warrant a narrower label once its subtree reveals it is really about venue costs — but surfacing it unobtrusively remains an open design problem.

5.2 Limitations

The current work is a design-and-build contribution without a formal user study. Mode-awareness operates at the prompt level only: modes reshape the AI's questioning style but not the structural form of the tree. The Plan Pane auto-refreshes on every answered question, which may produce stochastic variation across similar tree states, making it difficult for users to maintain a stable document; automatic refreshes bypass the accept/reject preview flow that otherwise protects user edits. Finally, recontextualization lacks a strong affordance, meaning one of the system's more sophisticated behaviours may go largely unnoticed.

5.3 Future Evaluation

The first evaluation we would conduct is a between-subjects comparison of midWife against a standard LLM chat interface on a structured open-ended planning task. The primary measure would be *decision coverage* — the number of distinct planning dimensions addressed, rated against a gold-standard checklist — directly operationalising midWife's central claim that Socratic tree-structured dialogue surfaces more of the problem space than linear chat. A between-subjects design avoids carryover effects from one tool's framing of the problem.

Secondary measures would include NASA-TLX, self-reported confidence in the plan, and time-on-task. A sample of 16–20 participants per condition would provide adequate power, consistent with prior work at this scale [15, 16].

5.4 Future Directions

Several natural extensions of midWife stand out. **Adaptive scaffolding** would adjust question depth and specificity based on observed user behaviour — how confidently they answer, how often they open the chat, how aggressively they elaborate — rather than relying solely on the selected mode. **Collaborative sessions** would extend the discourse tree to multiple users co-constructing a plan, which may be valuable for team decision-making and project scoping. **Richer export** — Gantt charts, structured task lists, or slide outlines generated directly from the tree — would allow the planning artifact to travel into a user's broader workflow. Finally, a **longitudinal study** could examine whether users return to and revise completed trees over time, and whether Socratic structure during planning produces better execution outcomes downstream.

6 Conclusion

Complex planning tasks routinely outgrow what a linear chat interface can hold. midWife addresses this by modelling deliberation as an interactive discourse tree in which an LLM acts as a Socratic interlocutor — asking rather than answering, probing rather than summarising — and distilling the result into an actionable Plan. The prototype demonstrates the feasibility of this paradigm across nine planning modes with a mixed-initiative interaction model. The natural next step is a between-subjects evaluation against standard LLM chat, testing whether Socratic tree-structured dialogue surfaces more of the problem space. We hope midWife opens a line of inquiry into interfaces that think *with* their users rather than merely respond to them.

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References

- [1] Edward Y. Chang. 2023. Prompting Large Language Models With the Socratic Method. arXiv:2303.08769
- [2] Chun-Wei Chiang, Zhuoran Lu, Zhuoyan Li, and Ming Yin. 2024. Enhancing AI-Assisted Group Decision Making through LLM-Powered Devil's Advocate. In *Proceedings of the 29th International Conference on Intelligent User Interfaces (IUI '24)*. ACM. doi:10.1145/3640543.3645207
- [3] Katherine M. Collins, Ilia Sucholutsky, Umang Bhatt, et al. 2024. Building Machines that Learn and Think with People. *Nature Human Behaviour* (2024). doi:10.1038/s41562-024-01991-9
- [4] Eric Horvitz. 1999. Principles of Mixed-Initiative User Interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '99)*. ACM, 159–166. doi:10.1145/302979.303030
- [5] Peiling Jiang, Jude Rayan, Steven P. Dow, and Haijun Xia. 2023. Graphologue: Exploring Large Language Model Responses with Interactive Diagrams. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM. doi:10.1145/3586183.3606737
- [6] Werner Kunz and Horst W. J. Rittel. 1970. *Issues as Elements of Information Systems*. Technical Report Working Paper 131. Institut für Grundlagen der Planung, Universität Stuttgart.
- [7] Hao-Ping Lee, Advait Sarkar, Lev Tankelevitch, et al. 2025. The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM. doi:10.1145/3706598.3713492
- [8] Jiayu Liu et al. 2024. SocraticLM: Exploring Socratic Personalized Teaching with Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS '24)*.
- [9] Michael Xieyang Liu, Tongshuang Wu, Tianying Chen, Franklin Mingzhe Li, Aniket Kittur, and Brad A. Myers. 2024. Selenite: Scaffolding Online Sensemaking with Comprehensive Overviews Elicited from Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM. doi:10.1145/3613904.3642149
- [10] Shuai Ma, Qiaoyi Chen, Xinru Wang, Chengbo Zheng, Zhenhui Peng, Ming Yin, and Xiaojuan Ma. 2025. Towards Human-AI Deliberation: Design and Evaluation of LLM-Empowered Deliberative AI for AI-Assisted Decision-Making. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM. doi:10.1145/3706598.3714021
- [11] Qiyu Pan et al. 2025. AMQuestioner: Training Critical Thinking with Question-Driven Interactive Argument Maps in Online Discussion. In *Proceedings of the ACM on Computer-Supported Cooperative Work and Social Computing (CSCW '25)*. ACM. doi:10.1145/3710926
- [12] Plato. 1990. *Theaetetus*. Hackett Publishing Company, Indianapolis, IN. Original work circa 369 BCE.
- [13] Horst W. J. Rittel and Melvin M. Webber. 1973. Dilemmas in a General Theory of Planning. *Policy Sciences* 4, 2 (1973), 155–169. doi:10.1007/BF01405730
- [14] Donald A. Schön. 1983. *The Reflective Practitioner: How Professionals Think in Action*. Basic Books.
- [15] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM. doi:10.1145/3613904.3642400
- [16] Sangho Suh, Bryan Min, Srishti Palani, and Haijun Xia. 2023. Sensecape: Enabling Multilevel Exploration and Sensemaking with Large Language Models. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM. doi:10.1145/3586183.3606756
- [17] Thitaree Tanprasert, David Okundaye, Anthony J. Perritano, and Ricarose Roque. 2024. Debate Chatbots to Facilitate Critical Thinking on YouTube: Social Identity and Conversational Style Make A Difference. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM. doi:10.1145/3613904.3642805 Best Paper Award.
- [18] Stephen E. Toulmin. 1958. *The Uses of Argument*. Cambridge University Press.
- [19] Lifei Wang et al. 2025. Evaluating Node-tree Interfaces for AI Explainability. arXiv:2503.04876
- [20] Zheng Zhang, Jie Gao, Ranjodh Singh Dhaliwal, and Toby Jia-Jun Li. 2023. VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM. doi:10.1145/3586183.3606746